Medical Physics
2nd Class
Physiology Lab **3**
Dr. Noor Mazin

Blood groups and Rh factor

Student Objectives:

- Define the "blood types" and the Rh factor, and the importance of blood grouping.
- Determine blood groups by using commercially available anti-sera, and precautions to be observed.



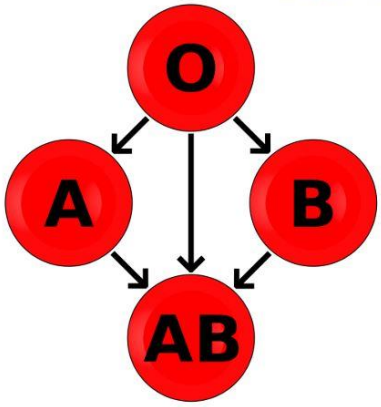
Introduction and Principle:

The surfaces of red cell membrane contain a variety of genetically determined antigens or (agglutinogens), while the plasma contains antibodies (agglutinins) that can react with these antigens of the RBCs surface when the two are mixed, this reaction called agglutination.

The blood group (type) of a person is determined by genes, these genes can be any one of 3 types—A, B, or O. The O gene is functionless and does not produce O antigen on red cells, while A and B genes produce strong antigens on red cells.

In addition to antigens of ABO system, the red cells of 80–85% of humans also contain an additional antigen, called Rh antigen or Rh factor or antigen D (Rh system) . Persons whose red cells contain this additional antigen are called "Rh positive" (Rh +) while those who lack this antigen are called "Rh negative" (Rh –) and this called **Rhesus (Rh) Factor**, however, approximately 3 out of every 20 people lack these antigens and are described as "Rh-negative" (Rh-).

To determine the blood group of a person, the red cells are made to react with commercially available antisera containing known agglutinins. The slide is then examined under the microscope to detect the presence or absence of clumping and hemolysis (agglutination) of red cells which occurs as a result of antigen-antibody reaction**.**



The importance of blood grouping:

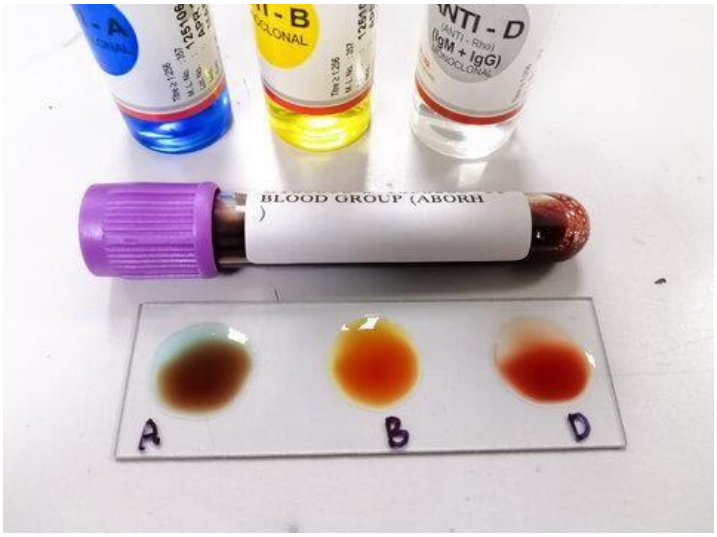
- In Blood transfusion for treatment purposes
- In Determination of Rh incompatibility between the mother and child.
- Choice of a donor in tissue/organ transplantation.
- Genetic studies.
- Susceptibility to disease.

Materials and instruments: **(slide method)**

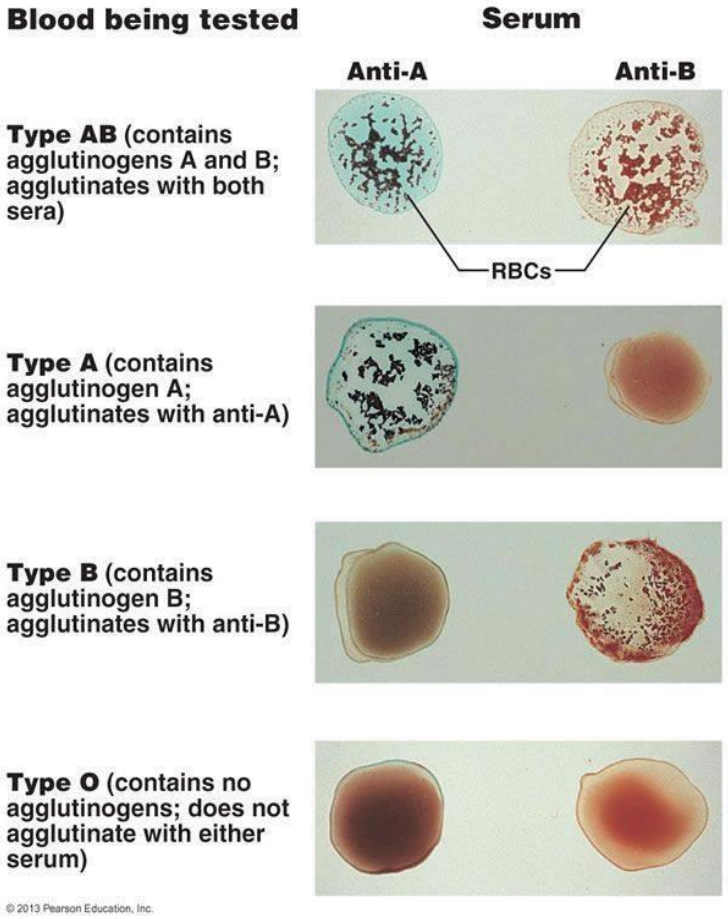
- Anti sera (anti-A, anti-B, and anti-D.
- Slides.
- Microscope.
- Applicator sticks for mixing.

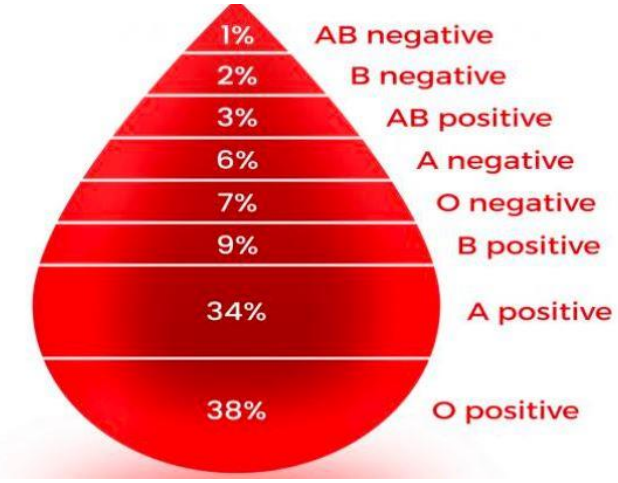
Procedure:

- Place three drops of blood on the slide.
- Then add one drop of each antibody (anti-A, anti-B, and anti-D) on each drop of blood.
- Mix with an applicator stick, wait for three minutes, and read the result with the naked eye.
- Examine under the microscope in case of confusion.
- The ABO and Rh blood types are usually distinguished together as in the figure below









Distribution of different blood types

	Group A	Group B	Group AB	Group O
Red blood cell type	A	B	AB	O
Antibodies present	Anti-B	Anti-A	None	Anti-A and Anti-B
Antigens present	A antigen	B antigen	A and B antigens	None

ABO System for Blood Group